

Web-Based Calculator - Pseudocode

What does a calculator do?

Let's first break down how a calculator works from a user's perspective:

1. I have a calculation in mind involving two numbers. 10×2400
2. Turn the calculator on with the "C" button and the display should turn on.
3. The process with a calculator looks like the following:
 - a. **First number input.**
 - i. The "1" button.
 - ii. The "0" button.
 - b. **Operator input.**
 - ii. The "x" button.
 - c. **Second number input.**
 - i. The "2" button.
 - ii. The "4" button.
 - iii. The "0" button.
 - iv. The "0" button.
 - d. **The calculation input.**
 - i. The "=" button.
 - e. **Results are displayed.** 24000 should be shown.

How does a calculator work?

Here's what goes into making each of those steps work:

1. Once the calculator is turned on, it is waiting for the first input.
2. The first number button is pressed. This number is now displayed.
3. The calculator, displaying the first integer entered for the first number, is now waiting for another integer or decimal point to continue the first number.
4. Once the first number is fully entered, the next step will begin once the user presses an operator button.
5. Once the operator button is selected, the calculator begins to wait for the start of the second number.
6. Once the second number is entered, the calculator will display the solution to the expression after the user presses the "=" button.

Variables

This is a pseudocode explanation of the variable process:

1. Upon Start

```
displayValue = "0" // the number that is currently displayed on the screen.=
firstNumber = null // the currently stored set of digits that constitutes the first number of the
expression
let operator = null // the field where the operator input of choice will be stored
inputNextNumber = false // whether or not to begin recording the second number (if integer
button = pressed)
```

2. The First Number Entered

```
currentDisplay = "10" // what is on the user's display
firstNumber = null // the first number is not stored until operator selected
operator = null // hasn't seen an operator yet
inputNextNumber = false
```

3. Operator Selection

```
currentDisplay = "10" // current user inputted values for first number
firstNumber = 10 // the number is now stored due to operator selection
operator = "x" // operation button pressed
inputNextNumber = true // now ready for second number
```

4. Next Number Entry

```
currentDisplay = "2400" // second number entered
firstNumber = 10 // stored first number
operator = "x" // stored operator value
inputNextNumber = true // shows second number entry enabled
```

5. When "=" is pressed

```
currentDisplay = "24000" // result displayed to user
firstNumber = 24000 // the result becomes the new first number
operator = "x" // stored operation to repeat the previous expression
inputNextNumber = true // ready for the next number
```

Actions to Functions

Each unique action determines a function of the calculator.

1. Number input - > inputDigit("1")
2. Operator input -> inputOperator("x")
3. Equals input - > inputEquals()

The Math

When the "=" is pressed:

let firstNumber = "inputDigit" // defines the first number inputs as the digits shown before operator choice

let currentDisplay = "inputDigit" // defines the next number inputs

case "x" // describes the type of operation function

return firstNumber * currentDisplay // executes the expression

When a new operator is selected:

```
switch (operator) {
  case '+': // if addition
    result = firstNumber + currentDisplay;
    break;
  case '-': // if subtraction
    result = firstNumber - currentDisplay;
    break;
  case '/': // if division
    result = firstNumber / currentDisplay;
```

User Input Function Example

If the user presses a number button:

```
function inputDigit(digit) { // defining the function
```

```
  if (currentDisplay === "0") { // if the display is blank...
```

```
    currentDisplay = digit; // show just the digit that was just inputted
```

```
  } else { // if there is already digits...
```

```
    currentDisplay += digit; // append the new digit to the already entered digits
```

```
  }
```

```
  updateDisplay(); // calls function to show changes on the display
```

Questions

What events do you need to use?

onClick = Number inputs. Operator inputs. Equals inputs. Clear inputs.

What functions will you write, and what will they do?

inputDigit(digit) – Number inputs

inputOperator(operator) – Operator selection

inputEquals() - calculates and displays results

updateDisplay() - shows changed results

How will you store data?

displayValue and **currentDisplay**, **firstNumber**, **operator**, **inputNextNumber** each store different values retrieved from functions.

To see this calculator in action, it can be downloaded and opened from the following link:

<https://github.com/x-alted/x-alted-scripts/blob/main/Web-Based-Calulator.html>